Heliyon 12 (2026) e44280

Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

Research article

# The adaptive markets hypothesis through the lens of machine learning

Julio Villavicencio [a,*], Oscar Miranda [b], Edwin Villanueva [b]

[a] Department of Economics, Pontifical Catholic University of Peru, 1801 Universitaria Avenue, San Miguel, Lima 15088, Peru
[b] Department of Engineering, Pontifical Catholic University of Peru, 1801 Universitaria Avenue, San Miguel, Lima 15088, Peru

ARTICLE INFO

*JEL classification:*
C12
C14
G14
G15

*Keywords:*
Adaptive markets hypothesis
S&P500
Deep learning
Machine learning
LSTM
Artificial intelligence

ABSTRACT

The adaptive markets hypothesis (AMH) postulates that markets are not stationary environments permanently in equilibrium as the efficient market hypothesis (EMH) posits, but evolve over time. However, testing this requires tools that capture the inherent complexity of financial markets. Therefore, in this paper we introduce an approach that uses the Long Short-Term Memory (LSTM) model to capture changes in the structure of the economic-financial environment. We apply the methodology to the U.S. equity market represented by the S&P500 for 40 years of daily data. With this tool we will demonstrate that, even with non-heuristic approaches such as the LSTM model, the learning and adaptation process after a change in the environment is slow, which is consistent with the AMH.

## 1. Introduction

The adaptive markets hypothesis (AMH) proposed by Lo [1,2] conjectures that markets are not stationary environments nor are they always in equilibrium as considered in the efficient markets hypothesis (EMH). This is because agents are not always rational and arrive at approximately optimal solutions by experimenting with heuristic methods. Then, when the environment changes such heuristics are no longer useful and mistakes are made. Finally, agents learn and adapt to the new environment, arriving at new solutions (see [2]). These processes cause financial markets to evolve permanently, turning them into complex dynamic systems that are difficult to analyze with non-dynamic approaches.

The most widely used approach to test the AMH is the variance ratio (VR) test of Lo and MacKinlay [3], which was initially used to test the random walk hypothesis, and later the efficient markets hypothesis. The VR test is based on the statistical property that if the price of an asset $p_t$ follows a random walk, then the variance of the $k$-period return is equal to $k$ times the variance of the one-period return. Due to its importance, since the publication of the original paper several authors have proposed improvements to the VR test (see [4–8]). However, we can highlight three challenges faced by VR-based tests in assessing AMH. First, VR tests assess the predictability of the market rather than whether there was a change in the structure of the environment in which agents operate. It could be the case that a radical change in the environment leads to high uncertainty about the future fundamentals of an equity market making its returns unpredictable. In this case the test would say that the market remained efficient because it remains unpredictable, at a time

* Corresponding author.
*Email address:* villavicencio.j@pucp.edu.pe (J. Villavicencio).

https://doi.org/10.1016/j.heliyon.2025.e44280
Received 6 March 2024; Received in revised form 18 November 2025; Accepted 24 November 2025
Available online 5 December 2025
2405-8440/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

when agents are more likely to make suboptimal decisions, leading to market inefficiency.[1] Secondly, the VR tests assess linear predictability. However, financial variables have complex underlying processes (hardly linear), with changing parameters over time, and cross correlation with other assets [see 10]. Finally, the VR tests tell us nothing about the variables associated with market dynamics.

Beyond the VR test, empirical applications that seek to test the EMH or the AMH have been incorporating other statistical tests to evaluate the linear and nonlinear dependence of the returns of stocks, indices, commodities, and cryptocurrencies, among other asset classes (see [9,11–15]). A widely used linear test, in addition to the VR test, is the portmanteau test proposed by Ljung and Box [16] and its subsequent extension such as the one developed by Escanciano and Lobato [17], as well as runs test and autocorrelation test among others. As for tests that adopt nonlinear measures of dependence, we can cite the generalized spectral test proposed by Escanciano and Velasco [18] and the consistent test by Dominguez and Lobato [19], among others.[2] All these tests, such as the VR test, are still based on statistical properties linked to the predictability (linear or non-linear) of the financial series, rather than addressing whether the underlying dynamics of the economic-financial environment are actually changing.

For the above reasons, in this research we seek to test the hypothesis of adaptive markets based on their own nature, that is, as the evolution of the economic-financial environment in the face of a shock that alters it. To achieve this, we first model the underlying market structure that governs the observed pattern of asset returns. Next, we employ statistical methods to detect significant changes in this pattern, indicating potential market regime shifts. Finally, we assess the market's adaptive capacity under two scenarios: (1) a passive adaptation where the market adjusts based on its prior knowledge of the structure, and (2) an active learning scenario where the model incorporates new information to refine its understanding of the post-shock environment.

To meet this challenge, it is necessary to opt for methods that make it possible to understand and recognize the evolution in the structure of the complex economic-financial environment in which market agents operate, by detecting changes in financial market patterns. The universal approximation theorem proposed by Cybenko [21], and its later extensions (see, for example,[22]), shows that any function or complex system can be "approximated" by an artificial neural network (ANN). In this line, Mustafa et al. [23] propose the use of a multilayer perceptron (MLP) neural network. However, such a kind of model does not have the natural ability to represent dynamic temporal processes. Therefore, we use the Long Short-Term Memory (LSTM) recurrent neural network to detect changes in market patterns (see [24,25]). LSTMs can capture both short- and long-term dependencies in market dynamics through their memory cells, allowing for better detection of genuine structural changes versus temporary anomalies. Moreover, their ability to selectively remember or forget information makes them particularly well-suited for studying market adaptation processes. This enables us to examine whether, even with sophisticated machine learning tools rather than simple heuristics, market adaptation remains a gradual process - a key implication of the AMH that has not been previously tested. By focusing on the evolutionary nature of markets through the lens of temporal deep learning, our methodology provides new insights into the fundamental mechanisms underlying market adaptation. Additionally, it should be noted that models such as the MLP and LSTM are flexible in the sense that they do not require a prior assumption of a linear or non-linear type of relationship of the time series, nor do they require a specific type of distribution.

We defined the economic-financial environment as the relationship between the returns of the stock market to be studied and the variables that underlie it. To demonstrate the feasibility of our approach we study the U.S. equity market,[3] represented by the returns of the S&P500 index over a 40-year daily sample.[4] To identify evolution in the environment we first train the model and then test its ability to reproduce market patterns. Therefore, we define changes in the environment as the situation in which the model loses the ability to explain the patterns of future return movements, measured by the drop in predictive performance, given that the economic-financial structure has changed.

This paper contributes to the literature in four ways. First, we introduce the deep learning LSTM recurrent network method to analyze the adaptive market hypothesis. To the best of our knowledge, this is the first time that there is an attempt to use these methods in the AMH literature. Secondly, this research demonstrates that even with sophisticated machine learning models like LSTMs, market adaptation to structural changes remains gradual, aligning with the core tenet of the adaptive markets hypothesis. More importantly, this result is a consequence of the nature of the learning process. Third, we explicitly consider explanatory variables that allow us to define the environment where the market evolution will take place. Finally, methodologically, we extend the work of Mustafa et al. [23] by showing that LSTM models are superior to MLPs as a suitable tool for detecting changes in the patterns of market returns because they consider the temporality inherent in the time series. The use of the LSTM model and the increase in the sample size can minimize the possibility of confusing a change in structure with a change in a variable belonging to the system that remained constant during training, as in the case of Mustafa et al. [23] with the Federal Reserve (Fed) reference rate.

We organize the paper as follows: besides this introduction, Section 2 discusses the data, variable definitions, and the methodology; Section 3 discusses the results; and Section 4 concludes.

## 2. Data and methodology

We propose leveraging the LSTM architecture to assess AMH. This is a type of recurrent neural network designed to capture short- and long-term dependencies in sequential data. Fig. 1 illustrates the internal mechanics of an LSTM unit. This processes a time series

[1] Kim et al. [9] find that during market crashes no significant return predictability is observed.

[2] See [20] for a comparative analysis of linear and nonlinear tests.

[3] This market was considered because it is the one where there is a high flow of information, which poses a greater challenge for testing adaptive markets. Further work could extend this methodology to other markets.

[4] We use daily data because it allows us to capture very short-term dynamics, such as the market reaction to a monetary policy decision. In addition, the granularity of the daily data allows us to have a large training sample, giving robustness to the results. Future work could apply it to lower frequencies such as weekly and monthly.

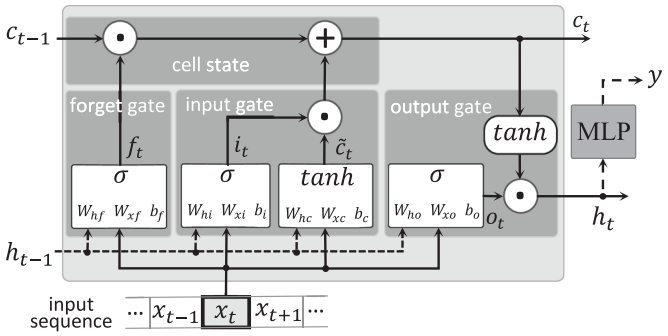



**Fig. 1. LSTM architecture**. At each time step $t$, the LSTM takes the input $x_t$ and the previous hidden state $h_{t-1}$, updating its memory cell $c_t$. The forget gate determines how much of the previous cell state $c_{t-1}$ should be retained, computed as: $f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$, where $\sigma(\cdot)$ is the sigmoid activation function. The input gate controls how much new information should be added to the cell state: $i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$. A candidate cell update $\tilde{c}_t$ is computed as: $\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$ ($\tanh(\cdot)$ is the hyperbolic tangent function). The new cell state $c_t$ is updated as a combination of the previous state and the candidate update: $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$ ($\odot$ represents element-wise multiplication). The output gate determines the next hidden state as $h_t = o_t \odot \tanh(c_t)$, where $o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$. All weight matrices $W$ and bias vectors $b$ are learnable parameters in training.

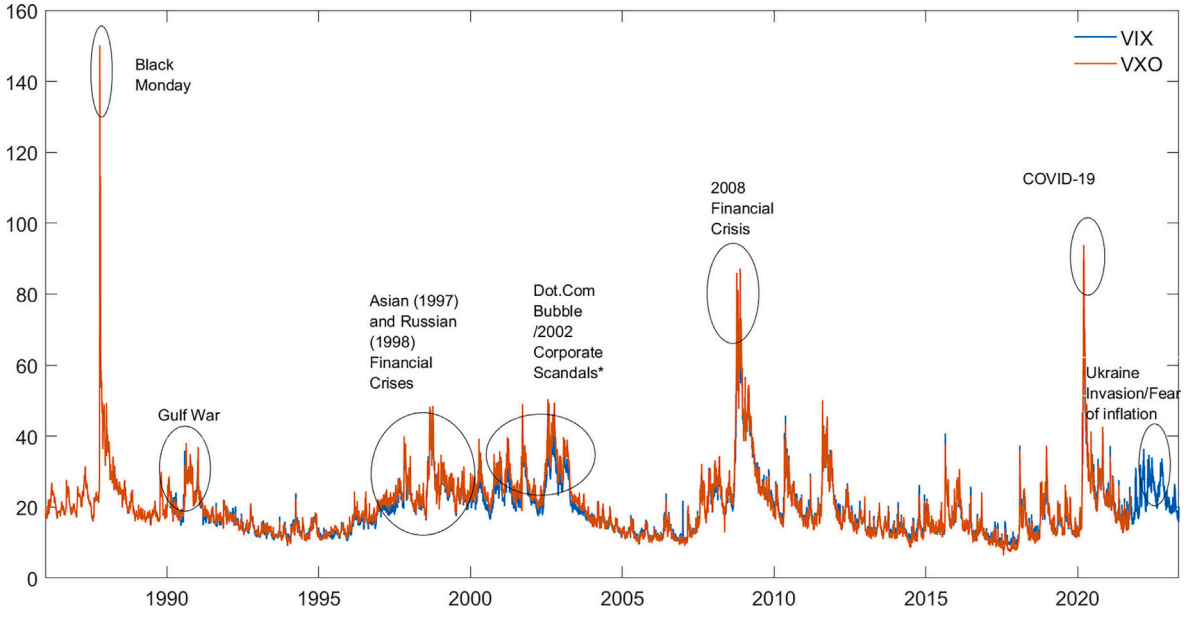



**Fig. 2.** VIX index and VXO index*.

* Corporate scandals refer to the period of major corporate accounting frauds, the largest of which was WorldCom in 2002.

by taking one time step $x_t$ at a time, updating at each step its memory cell $c_t$ and the hidden state $h_t$. In our approach, the final hidden state vector, obtained after processing the entire input sequence, is fed into a Multilayer Perceptron (MLP) network, which generates the output $y$.

In our approach, an input sequence consists of $K$ daily time points $[x_{t-K}, \ldots, x_{t-1}]$, where each $x_i$ is a vector of fundamental market variables at time $i$. The output $y$ represents the future return of the market index after the input sequence (at time $t$). To train and evaluate our models, we used daily data from the S&P 500 spanning April 14, 1983, to April 13, 2023, a period that includes significant high-uncertainty events (see Fig. 2).

We trained models $R$ to predict the future return of the S&P 500 index based on the previous $K$ time points of input variables, formally expressed as $y = R(\mathbf{X})$ with $\mathbf{X} = [x_{t-K}, \ldots, x_{t-1}]$. The input variables capture market expectations, monetary conditions, and external shocks, including the dollar price, oil price, money supply (M2), the Federal Reserve's effective rate, and U.S. Treasury bond yields at medium (5-year), long (10-year), and very long (30-year) maturities. The inclusion of Treasury yields across different maturities enables the model to account for term structure effects, which have been shown to predict business cycles (see [26–28]).

To integrate variables that are of lower frequency than daily (money supply M2 and Fed's effective rate) we use the stepwise integration methodology: the lower-frequency series are upsampled to match the highest frequency available (daily). This is done by holding the lower-frequency values constant over the corresponding higher-frequency periods until a new data point is available.

**Table 1**
Summary Statistics: daily data from April 14, 1983 to April 13, 2023.

| | S&P500 return | OIL return | US 5-year Bond Yield | US 10-yearBond Yield | US 30-year Bond Yield | Effective Fed Funds Rate | Log real Balance (M2) | Dollar return |
|---|---|---|---|---|---|---|---|---|
| Mean | 0.032 | 0.015 | 4.660 | 5.143 | 5.599 | 3.640 | 8.722 | −0.002 |
| Median | 0.029 | 0.000 | 4.444 | 4.689 | 5.120 | 3.250 | 8.699 | 0.000 |
| Maximum | 10.957 | 31.963 | 13.827 | 13.950 | 13.904 | 11.750 | 9.985 | 2.995 |
| Minimum | −22.900 | −40.048 | 0.191 | 0.508 | 0.997 | 0.125 | 7.642 | −3.546 |
| Std. Dev. | 1.130 | 2.476 | 2.976 | 2.811 | 2.586 | 3.021 | 0.646 | 0.515 |
| Skewness | −1.182 | −0.498 | 0.595 | 0.660 | 0.684 | 0.427 | 0.268 | −0.105 |
| Kurtosis | 29.389 | 23.517 | 2.685 | 2.861 | 2.985 | 2.097 | 1.926 | 5.229 |
| Jarque-Bera | 305,173 | 183,431 | 658 | 767 | 814 | 672 | 626 | 2180 |
| Probability | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| Observations | 10,434 | 10,434 | 10,434 | 10,434 | 10,434 | 10,434 | 10,434 | 10,434 |

Previous AMH studies typically rely only on the target variable (returns). Our approach incorporates a richer information set that helps identify genuine structural changes versus changes in individual variables. Table 1 shows the main statistics of the variables used. Financial variables tend to have non-normal distributions, as they contain extreme events. Therefore, it is observed that the returns show skewness coefficients different from zero and that the kurtosis is above 3, a value corresponding to non-normal distributions. This is corroborated by the Jarque-Bera test which rejects the null hypothesis of normality in the series. It should be noted that the LSTM model used in this research does not require the specification of a specific distribution of the time series.

Initially, in our first step, we follow the methodological strategy used by Mustafa et al. [23]. We locate the possible points where there may be a change in the structure. In the second stage, we seek to corroborate whether the change was really in the structure of the economic-financial environment.

To train the models, we generated a set of training samples $\{(y, \mathbf{X})\}_{t=1}^{T}$ from the T-length time series, where each sample $(y, \mathbf{X})$ consists of the return $y$ at time $t$ and the input sequence $\mathbf{X}$, which corresponds to the preceding $k$-length window. Checkpoints were established every fortnight (10 business days) throughout the period. For each checkpoint $p \in \{p_1, p_2, p_3, \ldots, p_p\}$, where $p_j = (j-1) \times 10 + K$ is the time index of the data, we train the LSTM model $R_p$ with the data for $S$ years prior to $p$, $K = 252 \times S$. For this study we experimented with two input windows, one of one year as in Mustafa et al. [23] and another of 10 years ($S = 1$ and 10). The latter window is close to the recent U.S. cycles which would allow the model to learn from the dynamics of the economic environment in both expansions and contractions[5] Thus we will define the training as $D_{train}(p) \equiv \{(y, \mathbf{X})\}_{t=p-K+1}^{t=p}$. With the trained model $R_p$, we test the data for the subsequent $M$ days after $p$, $D_{test}(p, M) \equiv \{(y, \mathbf{X})\}_{t=p+1}^{t=p+M}$. Given $M$, we construct a series of the root mean square errors, $\{RMSE(p, M)\}_{p=p_1}^{p_p}$, of the $R_p$ model trained with $D_{train}(p)$ and evaluated over $D_{test}(p, M)$[6]:

$$RMSE(p, M) = \sqrt{\frac{1}{M} \sum_{i=1}^{M} \epsilon_p(i)} \tag{1}$$

where $\epsilon_p(i) = \left(R_p\left(\mathbf{X}_{p+i}\right) - y_{p+i}\right)^2$ is the quadratic error for the day $i \in \{1, 2, \ldots, M\}$. If the structure changes at $j \in (p, p+M)$, the model trained with $D_{train}(p)$ fails to explain $D_{test}(p, M)$ which contains the data after the structural change $(j, p+M)$. This indicator will measure the degree of deviation of the new economic environment in which the agent has to operate from the training period. If the difference is large, then, at checkpoint $p$, a potential candidate for a change in the structure of the economic-financial environment has been found. We selected two values for $M$ equal to the number of working days in two and six months to differentiate short- and medium-duration events.

In the second stage, we will statistically test whether the events identified in the first stage represent a change in the economic-financial environment. To do so, we will analyze the behavior of the prediction errors before and after the high-uncertainty events, assuming that the model is not fed with the new information revealed. To identify whether the changes in the errors are statistically significant we applied the Ansari-Bradley test. This test evaluates the null hypothesis that the data in two vectors come from the same distribution, under the alternative that they come from the same median and shape but with different dispersion. In these first two stages we will compare the results of the LSTM model with those of the MLP used by Mustafa et al. [23].

Finally, once the existence of events representing changes in the environment has been proven statistically, the AMH will be tested under the assumption that the model learns daily from new information and adapts in line with it. If the market is adaptive it would be expected that, even using a non-heuristic tool such as the LSTM, the prediction errors do not immediately converge to the situation prior to the change in the environment, but that there is a period of convergence.

[5] According to the National Bureau of Economic Research, the average U.S. cycle, measured from peak to peak, between 1945 and 2020 has been 6.3 years. However, the last four cycles, which coincide with the sample used in this paper, have an average duration of 9.64 years. Therefore, we use 10 years as a base (see https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions).

[6] This is equivalent to the root of what Mustafa et al. [23] calls "structure-monitoring index". We decided to use the root so that the resulting measure can be comparable with the returns.

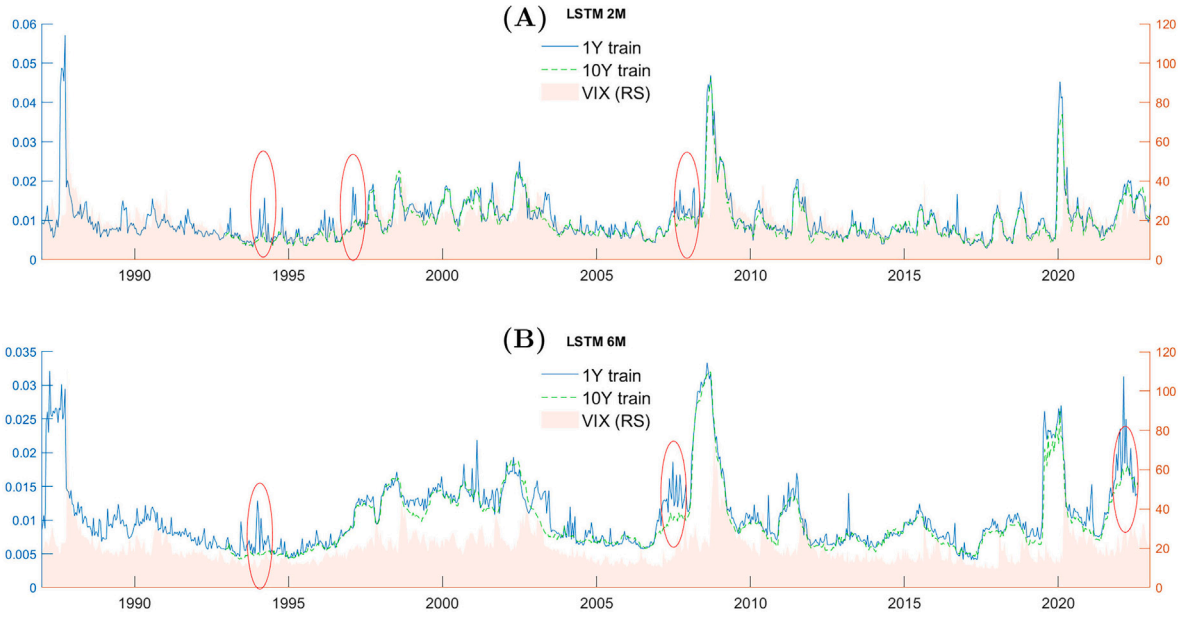



**Fig. 3.** RMSE of the LSTM model with 1 and 10 years of training. (A) Forecast horizon of 2 months. (B) Forecast horizon of 6 months. Notes: The VIX index is completed with the VXX index before January 1990. The shaded area represents the VIX index (right scale), and the lines show the RMSE of the LSTM model trained on 1 and 10 years of data.

## 3. Empirical results

### *3.1. Identifying candidates for changes in the economic-financial environment*

Figs. 3 and 4 show the RMSE of the LSTM and MLP models, respectively, for 1 and 10 years of training, when testing 2 and 6 months of data after each checkpoint.[7] The results are in line with the AMH in the sense that errors increase during periods of high uncertainty, such as those shown in Fig. 2, but then decrease gradually as new information is incorporated into the model.

However, there are important differences between the models shown in Table 2. First, LSTM models are superior to MLP models (lower RMSE) because they exploit the temporal structure of the information. This advantage allows the model to distinguish a change in the structure of the environment from a change in one of its component variables, since during its learning, the fact that it has memory allows it to remember this difference. As a result, for long training periods, the RMSE obtained from the LSTM model does not jump when faced with a change in one of the variables. In particular, unlike Mustafa et al. [23], the LSTM model with 10-year training used does not jump when there is an increase in the Fed's reference rate after a long period of immobility such as February 1994, March 1997 and March 2022 (marked with a red ellipse in the figures).

Second, as expected, models with longer training periods perform better. This is because, by having a longer record of past events, the model learns to better differentiate a change in the pattern of returns resulting from the evolution of the economic-financial structure from one resulting from changes in the explanatory variables. As expected, this improvement is greater in MLP models, since they do not recognize the temporal structure of the data. On the contrary, in an LSTM model, the most recent information could have, in some cases, a greater weight than the oldest one.

### *3.2. Statistical testing of candidates for changes in the economic-financial environment*

In this section we will test whether the candidates for changes in the environment are statistically significant. For that purpose, we will now evaluate whether the errors before and after the high-uncertainty events found in the previous section are statistically different, without incorporating the information that emerged during that period into our models. If the errors obtained after the event are permanently higher (even after the initial impact loses strength) we will interpret that the economic-financial environment has changed, causing a reduction in the model's ability to recognize market return patterns. Fig. 5 shows the prediction errors for the 280 days surrounding each of the uncertainty events identified in the previous section. To identify whether the changes in the errors are statistically significant, the Ansari-Bradley test is applied, which evaluates the null hypothesis that the data in two vectors come from the same distribution, under the alternative that they come from the same median and shape but different dispersion.

Table 3 shows the test results for two cases: when including (left side) and excluding the high-uncertainty event (right side). In general, under a significance level of 5 %, the test rejects the null hypothesis that the errors come from the same distribution. The

[7] The hyperparameters used for both models can be found in Appendix A.

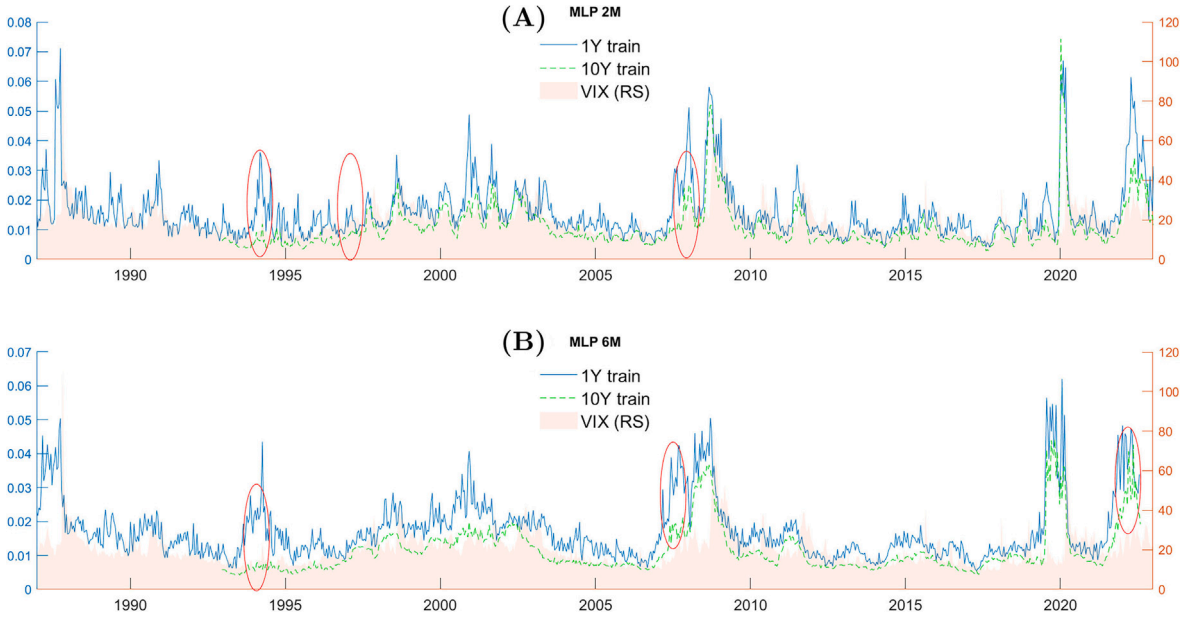



**Fig. 4.** RMSE of the MLP model with 1 and 10 years of training. (A) Forecast horizon of 2 months. (B) Forecast horizon of 6 months. Notes: The VIX index is completed with the VXX index before January 1990. The shaded area represents the VIX (right scale), and the lines show the RMSE of the MLP model trained on 1 and 10 years of data.

**Table 2**
RMSE and RMdSE for different horizons and models.

| | | MLP | | | | LSTM | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 1 Y Train | | 10 Y Train | | 1 Y Train | | 10 Y Train | |
| | | 2 M test | 6 M test | 2 M test | 6 M test | 2 M Test | 6 M Test | 2 M test | 6 M test |
| Average RMSE | 12/15/1992-1/3/2023 | 1.55 % | 1.71 % | 1.06 % | 1.14 % | 1.02 % | 1.08 % | 0.94 % | 0.99 % |
| | 4/3/1984-1/3/2023 | 1.64 % | 1.79 % | n.a. | n.a. | 1.02 % | 1.07 % | n.a. | n.a. |
| Average RMdSE | 12/15/1992-1/3/2023 | 1.27 % | 1.42 % | 0.84 % | 0.89 % | 0.85 % | 0.94 % | 0.78 % | 0.84 % |
| | 4/3/1984-1/3/2023 | 1.34 % | 1.51 % | n.a. | n.a. | 0.86 % | 0.92 % | n.a. | n.a. |

Notes: In the table we present the root mean squared errors (RMSE) and root median squared errors (RMdSE) of the MLP and LSTM models with 1 and 10 years of training. We consider out-of-sample prediction horizons of 2 and 6 months. They are separated into two samples because for the models trained with 10 years we estimate the prediction error since December 15, 1992.

exception is the Asian Financial Crisis and the Dot.com Bubble only for the case of the MLP model with 1 year of training. When we remove the event itself from the comparison samples, the results hold in general, with the exception of the Russian Financial Crisis for the MLP model with 10 years of training (although at 10 % confidence the null hypothesis is also rejected). As expected, models with a shorter training period have a weaker rejection of the null hypothesis, as is the case for the Dot.Com Bubble.

### *3.3. The model with adaptation*

In this section we will have the model learn daily from the new environment and adapt based on the information it acquires. The question is whether under these new assumptions the model can better cope with high-uncertainty events, converging to errors close to those prior to such events. Under AMH it would be expected that, despite learning about new information through non-heuristic methods such as LSTM, the adaptation process will be slow. Table 4 shows that the Ansari-Bradley tests are not very different from those obtained in the previous section (see Table 3), where the model was not fed with new information.

Therefore, it is important to study what happens with the daily learning and adaptation process, given that we are dealing with sophisticated machine learning models and not heuristic tools. Fig. 6 shows the dynamics of the prediction error of the LSTM models with 1 and 10 years of training during the 2008 Financial Crisis.[8] In each case we will consider the model with and without learning and adaptation. We observe that the errors after the event are still larger than before, despite the existence or not of learning and adaptation. More importantly, when the models used have a longer training period, the learning and adaptation process is slower. We interpret that this result occurs because the history ends up outweighing the new information in the model prediction, potentially revealing a weakness in the LSTM model. As a result, the models with and without learning and adaptation are very similar in the

[8] The results for the remaining events can be requested. The conclusions are similar to those explained for the 2008 Financial Crisis.

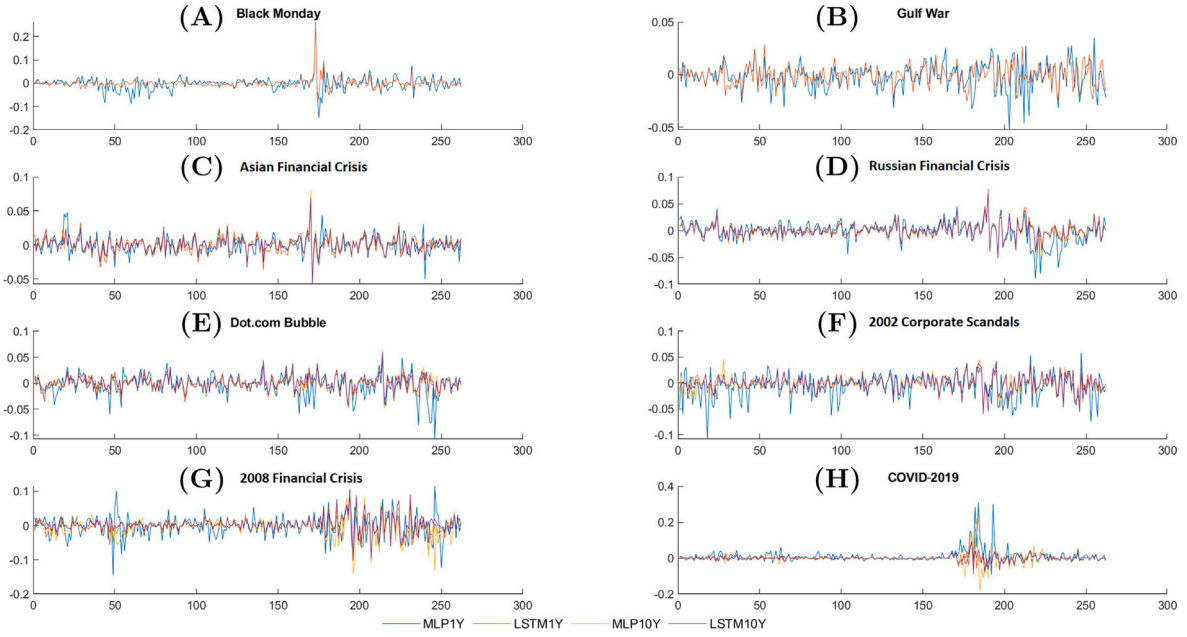



**Fig. 5.** Daily prediction errors of MLP and LSTM models across each uncertainty event, considering training periods of one year (1 Y) and ten years (10 Y). (A) Black Monday, (B) Gulf War, (C) Asian Financial Crisis, (D) Russian Financial Crisis, (E) Dot.com Bubble, (F) 2002 Corporate Scandals, (G) 2008 Financial Crisis, (H) COVID-19. The models do not incorporate or process new information during the displayed sample.

**Table 3**
Ansari-Bradley test p-value for model prediction errors without learning and adaptation.

| | Considering the event | | | | Without considering the event | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | MLP1Y | LSTM1Y | MLP10Y | LSTM10Y | | MLP1Y | LSTM1Y | MLP10Y | LSTM10Y |
| Black Monday | 0.00109 | 0.00000 | | | Black Monday | 0.01706 | 0.00004 | 0.00000 | 0.00000 |
| Gulf War | 0.00000 | 0.00001 | | | Gulf War | 0.00000 | 0.00001 | 0.00000 | 0.00000 |
| Asian Financial Crisis | 0.22668 | 0.59131 | 0.15219 | 0.08442 | Asian Financial Crisis | 0.66990 | 0.19300 | 0.70022 | 0.46332 |
| Russian Financial Crisis | 0.00000 | 0.00008 | 0.00010 | 0.00004 | Russian Financial Crisis | 0.00000 | 0.03871 | 0.08075 | 0.02652 |
| Dot.com Bubble | 0.06318 | 0.03569 | 0.00199 | 0.00100 | Dot.com Bubble | 0.12109 | 0.17040 | 0.00625 | 0.00490 |
| 2002 Corporate Scandals | 0.00102 | 0.00000 | 0.00000 | 0.00000 | 2002 Corporate Scandals | 0.04922 | 0.00007 | 0.00014 | 0.00002 |
| 2008 Financial Crisis | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 2008 Financial Crisis | 0.00000 | 0.00000 | 0.00000 | 0.00000 |
| COVID-19 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | COVID-19 | 0.00003 | 0.00000 | 0.00000 | 0.00000 |

Notes: Numbers shaded with red indicate non-rejection of the null hypothesis at 5 % confidence.

**Table 4**
Ansari-Bradley test p-value for model prediction errors with learning and adaptation.

| | Considering the Event | | | | Without considering the event | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | MLP1Y | LSTM1Y | MLP10Y | LSTM10Y | | MLP1Y | LSTM1Y | MLP10Y | LSTM10Y |
| Black Monday | 0.00003 | 0.00000 | | | Black Monday | 0.00462 | 0.00006 | | |
| Gulf War | 0.00348 | 0.00000 | | | Gulf War | 0.00108 | 0.00000 | | |
| Asian Financial Crisis | 0.00932 | 0.09527 | 0.02127 | 0.07134 | Asian Financial Crisis | 0.04184 | 0.38381 | 0.38381 | 0.56048 |
| Russian Financial Crisis | 0.00012 | 0.00251 | 0.00006 | 0.00003 | Russian Financial Crisis | 0.00119 | 0.17340 | 0.01438 | 0.03744 |
| Dot.com Bubble | 0.06217 | 0.04429 | 0.00050 | 0.00016 | Dot.com Bubble | 0.04545 | 0.11424 | 0.00139 | 0.00089 |
| 2002 Corporate Scandals | 0.00229 | 0.00000 | 0.00001 | 0.00000 | 2002 Corporate Scandals | 0.00380 | 0.00000 | 0.00118 | 0.00001 |
| 2008 Financial Crisis | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 2008 Financial Crisis | 0.00000 | 0.00000 | 0.00000 | 0.00000 |
| COVID-19 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | COVID-19 | 0.00004 | 0.00000 | 0.00000 | 0.00000 |

Notes: Numbers shaded with red indicate non-rejection of the null hypothesis at 5 % confidence.

days after the event and this similarity is greater in the model with 10 years of training than in the one with only 1 year of training. In conclusion, the strength of having a longer training period that allows lower prediction errors to be achieved in the total sample, ends up being a weakness as it reduces the speed of adaptation. Therefore, even using non-heuristic methods such as the LSTM machine learning model, in the face of an uncertainty event errors will increase, which is in line with the AMH even in this case.

## 4. Conclusions

In this paper, we seek to test the hypothesis of adaptive markets based on their very nature, which is the existence of an evolutionary process following a shock that alters the economic-financial environment that governs the pattern of returns. To this end, we

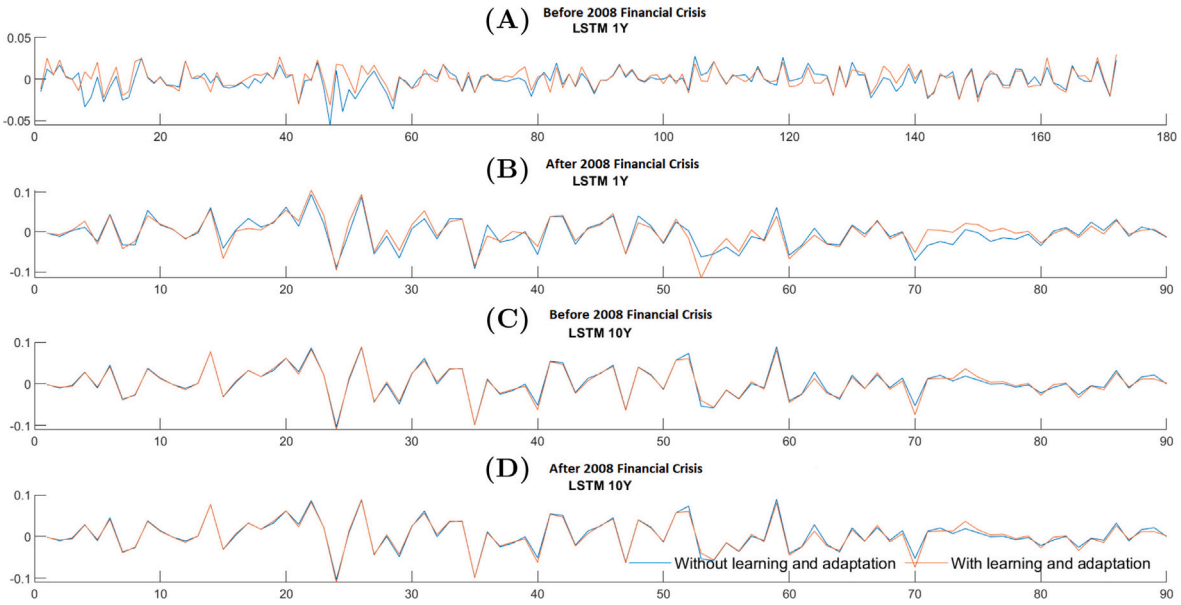



**Fig. 6.** LSTM prediction errors with and without learning and adaptation around the 2008 Financial Crisis. (A) Before the 2008 financial crisis – 1 Y training; (B) After the 2008 financial crisis – 1 Y training; (C) Before the 2008 financial crisis – 10 Y training; (D) After the 2008 financial crisis – 10 Y training. Models with and without learning/adaptation are compared in each panel.

employ a Long Short-Term Memory (LSTM) recurrent neural network to identify the structure underlying the pattern of returns and the changes it might undergo. LSTMs are designed to capture both short-term and long-term dependencies within market dynamics via their memory cells, making them effective for distinguishing between genuine structural changes and temporary fluctuations. Moreover, their ability to selectively remember or forget information makes them particularly well-suited for studying market adaptation processes. This approach allows us to assess whether, even with sophisticated machine learning models, market adaptation is inherently gradual, a core implication of the Adaptive Market Hypothesis (AMH).

We apply the methodology to the U.S. equity market represented by the S&P500 for 40 years of daily data. We find that when faced with an event that affects the economic-financial environment, the market adapts only gradually, regardless of whether it acquires and learns new information after the shock, which is in line with the AMH. This result is observed despite the fact that a non-heuristic approach such as the LSTM is being used for the adaptation process. More importantly, in our research the process of adaptability is explained by the use of memory in learning. Although memory has the advantage of allowing recall of historical information useful for the future, it also generates delays in learning new information after a change in the learned structure has occurred. As a consequence, the adaptation process will depend on how the new information, in the learning process, makes a space in the memory.

In addition, the LSTM model, by having memory, avoids interpreting a change in a variable that composes the system as a change in the structure of the system. As a consequence, we find, for example, that the LSTM model correctly interprets as a change in a variable and not in the system the increase in the Fed's benchmark interest rate after a long period of immobility in February 1994, March 1997 and March 2022. In contrast, the MLP model, which does not have the ability to represent temporal dynamics, interprets changes in rates as changes in structure at those dates, leading to a considerable increase in errors.

By focusing on the evolutionary nature of markets through the lens of temporal deep learning, our methodology provides new insights into the fundamental mechanisms underlying market adaptation. For investors, these results not only entail recognizing and understanding that markets are evolutionary, but also employing methodologies and frameworks that enable them to devise effective investment strategies in such dynamic environments. Artificial intelligence emerges as a potential ally in the creation of such solutions. For researchers, the results provide a foundation for future investigations into the evolutionary processes of different asset markets through machine learning techniques. As future research in this line, we recommend evaluating alternative learning processes that the artificial intelligence literature has been developing and extend the study to other markets such as emerging market assets, fixed income, among others.

## CRediT authorship contribution statement

**Julio Villavicencio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Oscar Miranda:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Edwin Villanueva:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

## Declaration of AI-assisted technologies in the writing process

During the preparation of this work the authors used DeepL in order to improve the language of the text. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

This project has received financial support from the Pontifical Catholic University of Peru (CAP project PI0762).

## Appendix A. Hyperparameters

The results shown in the paper originate from LSTM and MLP models that have the following hyperparameters.

| Name of hyperparameters | LSTM | ANN (MLP) |
|---|---|---|
| Number of time steps | 12 | 12 |
| Number of input layers | 1 | 1 |
| No of neurons in input layer | 7 | 7 |
| Number of hidden layers | 1 | 1 |
| No of neurons in hidden layer | 10 | 10 |
| Number of epochs required | 50 | 50 |
| Activation funcion in input layer (forward) | ReLU | ReLU |
| Activation funcion in hidden layer | ReLU | ReLU |
| Loss optimizer used | Adam | Adam |
| Dropout rate | 0 | 0,0.2 |

## Appendix B. Details of RMSEs before and after high uncertainty events

In Table 5 we present the detail of the RMSEs before and after for each event, with and without adaptation, and by training time.

**Table 5**
RMSE before and after the event: with and without adaptation.

| | Static LSTM | | | Dynamic LSTM | | | Static MLP | | | Dynamic MLP | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Before the event | After the event | Diff. | Before the event | After the event | Diff. | Before the event | After the event | Diff. | Before the event | After the event | Diff. |
| Black Monday 1 Y training | 0.07 % | 0.33 % | 0.26 % | 0.07 % | 0.33 % | 0.26 % | 0.18 % | 0.41 % | 0.23 % | 0.17 % | 0.55 % | 0.38 % |
| Gulf War 1 Y training | 0.06 % | 0.13 % | 0.07 % | 0.06 % | 0.15 % | 0.09 % | 0.08 % | 0.19 % | 0.11 % | 0.10 % | 0.23 % | 0.13 % |
| Asian Crisis 1 Y training | 0.09 % | 0.13 % | 0.04 % | 0.07 % | 0.16 % | 0.08 % | 0.10 % | 0.17 % | 0.07 % | 0.09 % | 0.19 % | 0.10 % |
| Asian Crisis 10 Y training | 0.07 % | 0.14 % | 0.07 % | 0.07 % | 0.15 % | 0.08 % | 0.07 % | 0.15 % | 0.07 % | 0.07 % | 0.15 % | 0.08 % |
| Russian Crisis 1 Y training | 0.07 % | 0.19 % | 0.12 % | 0.07 % | 0.17 % | 0.10 % | 0.10 % | 0.28 % | 0.19 % | 0.11 % | 0.22 % | 0.12 % |
| Russian Crisis 10 Y training | 0.07 % | 0.18 % | 0.11 % | 0.07 % | 0.18 % | 0.11 % | 0.07 % | 0.19 % | 0.12 % | 0.07 % | 0.19 % | 0.12 % |
| Dot.Com Bubble 1 Y training | 0.09 % | 0.19 % | 0.10 % | 0.09 % | 0.19 % | 0.09 % | 0.13 % | 0.32 % | 0.19 % | 0.13 % | 0.26 % | 0.13 % |
| Dot.Com Bubble 10 Y training | 0.08 % | 0.19 % | 0.11 % | 0.08 % | 0.19 % | 0.11 % | 0.09 % | 0.20 % | 0.11 % | 0.08 % | 0.19 % | 0.11 % |
| Corporate Scandals 1 Y training | 0.08 % | 0.22 % | 0.13 % | 0.09 % | 0.22 % | 0.13 % | 0.17 % | 0.32 % | 0.15 % | 0.15 % | 0.26 % | 0.11 % |
| Corporate Scandals 10 Y training | 0.09 % | 0.23 % | 0.15 % | 0.09 % | 0.23 % | 0.14 % | 0.10 % | 0.24 % | 0.14 % | 0.10 % | 0.23 % | 0.13 % |
| 2008 Crisis 1 Y training | 0.10 % | 0.38 % | 0.28 % | 0.09 % | 0.38 % | 0.29 % | 0.20 % | 0.48 % | 0.28 % | 0.18 % | 0.50 % | 0.32 % |
| 2008 Crisis 10 Y training | 0.08 % | 0.39 % | 0.31 % | 0.09 % | 0.39 % | 0.30 % | 0.12 % | 0.51 % | 0.38 % | 0.16 % | 0.49 % | 0.34 % |
| COVID 1 Y training | 0.06 % | 0.30 % | 0.24 % | 0.06 % | 0.29 % | 0.23 % | 0.11 % | 0.64 % | 0.53 % | 0.10 % | 0.58 % | 0.48 % |
| COVID 10 Y training | 0.05 % | 0.25 % | 0.20 % | 0.05 % | 0.27 % | 0.22 % | 0.07 % | 0.50 % | 0.43 % | 0.06 % | 0.37 % | 0.31 % |

## Data and code availability

Data will be made available on request. For data requests, please write to the corresponding author.

## References


[1] A.W. Lo, The adaptive market hypothesis: market efficiency from a evolutionary perspective, J. Portf. Manag. 30 (5) (2004) 15–29, https://doi.org/10.3905/jpm.2004.442611, https://jpm.pm-research.com/content/30/5/15.

[2] A.W. Lo, Reconciling efficient markets with behavioral finance: the adaptive markets hypothesis, J. Invest. Consult. 7 (2) (2005) 21–44.

[3] A.W. Lo, A.C. MacKinlay, Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test, Rev. Financ. Stud. 1 (1) (1988) 41–66. arXiv: https://doi.org/10.1093/rfs/1.1.41, https://academic.oup.com/rfs/article-pdf/1/1/41/24434733/010041.pdf.

[4] J.H. Wright, Alternative variance-ratio tests using ranks and signs, J. Bus. Econ. Stat. 18 (1) (2000) 1–9, http://www.jstor.org/stable/1392131.

[5] J.H. Kim, Wild bootstrapping variance ratio tests, Econ. Lett. 92 (1) (2006) 38–43, https://doi.org/10.1016/j.econlet.2006.01.007, https://www.sciencedirect.com/science/article/pii/S0165176506000140.

[6] K. Chow, K.C. Denning, A simple multiple variance ratio test, J. Econom. 58 (3) (1993) 385–401, https://doi.org/10.1016/0304-4076(93)90051-6, https://www.sciencedirect.com/science/article/pii/0304407693900516.

[7] J. Belaire-Franch, D. Contreras, Ranks and signs-based multiple variance ratio tests, in: Spanish-Italian Meeting on Financial Mathematics, VII, Spanish-Italian Meeting on Financial Mathematics Cuenca, 2004, pp. 40.

[8] S.Y. Hong, O. Linton, H.J. Zhang, An investigation into multivariate variance ratio statistics and their application to stock market predictability, J. Financ. Econom. 15 (2) (2017) 173–222, https://doi.org/10.1093/jjfinec/nbw014, https://academic.oup.com/jfec/article-pdf/15/2/173/11203139/nbw014.pdf.

[9] J.H. Kim, A. Shamsuddin, K.-P. Lim, Stock return predictability and the adaptive markets hypothesis: Evidence from century-long U.S. data, J. Empir. Finance 18 (5) (2011) 868–879, https://doi.org/10.1016/j.jempfin.2011.08.002, https://www.sciencedirect.com/science/article/pii/S0927539811000612.

[10] K.J. Forbes, R. Rigobon, Contagion, Only interdependence: measuring stock market comovements, J. Finance 57 (5) (2002) 2223–2261, https://doi.org/10.1111/0022-1082.00494, https://ideas.repec.org/a/bla/jfinan/v57y2002i5p2223-2261.html.

[11] M. Dos-Santos, L. Favero, T. Brugni, R. Serra, Adaptive markets hypothesis and economic-institutional environment: a cross-country analysis, Rev. Gest. 31 (2024) https://doi.org/10.1108/REGE-06-2022-0095, https://www.emerald.com/insight/content/doi/10.1108/rege-06-2022-0095/full/html.

[12] X. Brouty, M. Garcin, A statistical test of market efficiency based on information theory, Quant. Finance 23 (6) (2023) 1003–1018, https://doi.org/10.1080/14697688.2023.2211108

[13] M. Shahid, M. Islam, N. Alam, M. Ali, Time-varying return predictability and adaptive behavior in the u.s. commodity markets during covid-19, Int. J. Econ. Manag. 16 (S1) (2022) 59–80 ISSN 1823-836X. e-ISSN 2600–9390 https://doi.org/10.47836/IJEAMSI.16.1.005

[14] B. Bhuyan, S. Patra, R.K. Bhuian, Market adaptability and evolving predictability of stock returns: an evidence from india, Asia-Pac. Financ. Mark. 27 (4) (2020) 605–619.

[15] M.N. Shahid, S. Coronado, A. Sattar, Stock market behaviour: efficient or adaptive? Evidence from the Pakistan Stock Exchange, Afro-Asian J. Finance Account. 9 (2) (2019) 167–192, https://ideas.repec.org/a/ids/afasfa/v9y2019i2p167-192.html.

[16] G. Ljung, G. Box, On a measure of lack of fit in time series models, Biometrika 65 (Aug 1978) https://doi.org/10.1093/biomet/65.2.297

[17] J.C. Escanciano, I.N. Lobato, An automatic portmanteau test for serial correlation, J. Econ., 151 (2) (2009) 140–149, [Recent Advances in Time Series Analysis: A Volume Honouring Peter M. Robinson] https://doi.org/10.1016/j.jeconom.2009.03.001. https://www.sciencedirect.com/science/article/pii/S0304407609000773.

[18] J.C. Escanciano, C. Velasco, Generalized spectral tests for the martingale difference hypothesis, J. Econ. 134 (1) (2006) 151–185, https://doi.org/10.1016/j.jeconom.2005.06.019, https://www.sciencedirect.com/science/article/pii/S0304407605001417.

[19] M. Dominguez, I. Lobato, Testing the martingale difference hypothesis, Econom. Rev. 22 (2003) 351–377, https://doi.org/10.1081/ETC-120025895

[20] A. Charles, O. Darne, J.H. Kim, Small sample properties of alternative tests for martingale difference hypothesis, Econ. Lett. 110 (2) (2011) 151–154, https://doi.org/10.1016/j.econlet.2010.11.018, https://www.sciencedirect.com/science/article/pii/S0165176510003885.

[21] G. Cybenko, Approximation by superpositions of a sigmoidal function, Math. Control Signals Syst. 2 (4) (1989) 303–314.

[22] K. Hornik, Approximation capabilities of multilayer feedforward networks, Neural Netw. 4 (2) (1991) 251–257, https://doi.org/10.1016/0893-6080(91)90009-T, https://www.sciencedirect.com/science/article/pii/089360809190009T.

[23] A.A. Mustafa, C.-Y. Lin, M. Kakinaka, Detecting market pattern changes: a machine learning approach, Financ. Res. Lett. 47 (2022) 102621, https://doi.org/10.1016/j.frl.2021.102621, https://www.sciencedirect.com/science/article/pii/S1544612321005572.

[24] F. Salem, Recurrent Neural Networks: From Simple to Gated Architectures, Springer International Publishing, 2022, https://books.google.com.pe/books?id=bJpXEAAAQBAJ.

[25] S. Hochreiter, J. Schmidhuber, Long Short-Term Memory, Neural Comput. 9 (8) (1997) 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735, https://direct.mit.edu/neco/article-pdf/9/8/1735/813796/neco.1997.9.8.1735.pdf.

[26] A. Estrella, G.A. Hardouvelis, The term structure as a predictor of real economic activity, J. Finance 46 (2) (1991) 555–576, https://doi.org/10.1111/j.1540-6261.1991.tb02674.x, https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1991.tb02674.x.

[27] D. Sabes, J.-G. Sahuc, Do yield curve inversions predict recessions in the euro area? Financ. Res. Lett. 52 (2023) 103416, https://doi.org/10.1016/j.frl.2022.103416, https://www.sciencedirect.com/science/article/pii/S1544612322005931.

[28] J.L. Cendejas, Recessions and flattening of the yield curve (1960 - 2021): a two-way road under a regime switching approach, Q. Rev. Econ. Finance 88 (2023) 8–20, https://doi.org/10.1016/j.qref.2022.12.007, https://www.sciencedirect.com/science/article/pii/S1062976922001417.